11ra Clase 31/11/2025

75.43 Introducción a los Sistemas Distribuidos
95.60 Redes y Aplicaciones Distribuidas
TA048 Redes

Tema: Capa Física Inalámbrica (I)

Capítulo 7 hasta *7.3 WiFi: 802.11 Wireless LANs* inclusive; *Computer Networking: A Top-Down Approach. James Kurose and Keith Ross. Publisher: Pearson, Edition: 8th, 2021.*

Dr. Ing. J. Ignacio Alvarez-Hamelin

Clase de hoy

- LAN inalámbricas
- Características físicas de los medios
- WiFi: IEEE 802.11

LAN Inalámbricas

- Tecnologías según alcance
- Necesidades (infraestructura, movilidad)

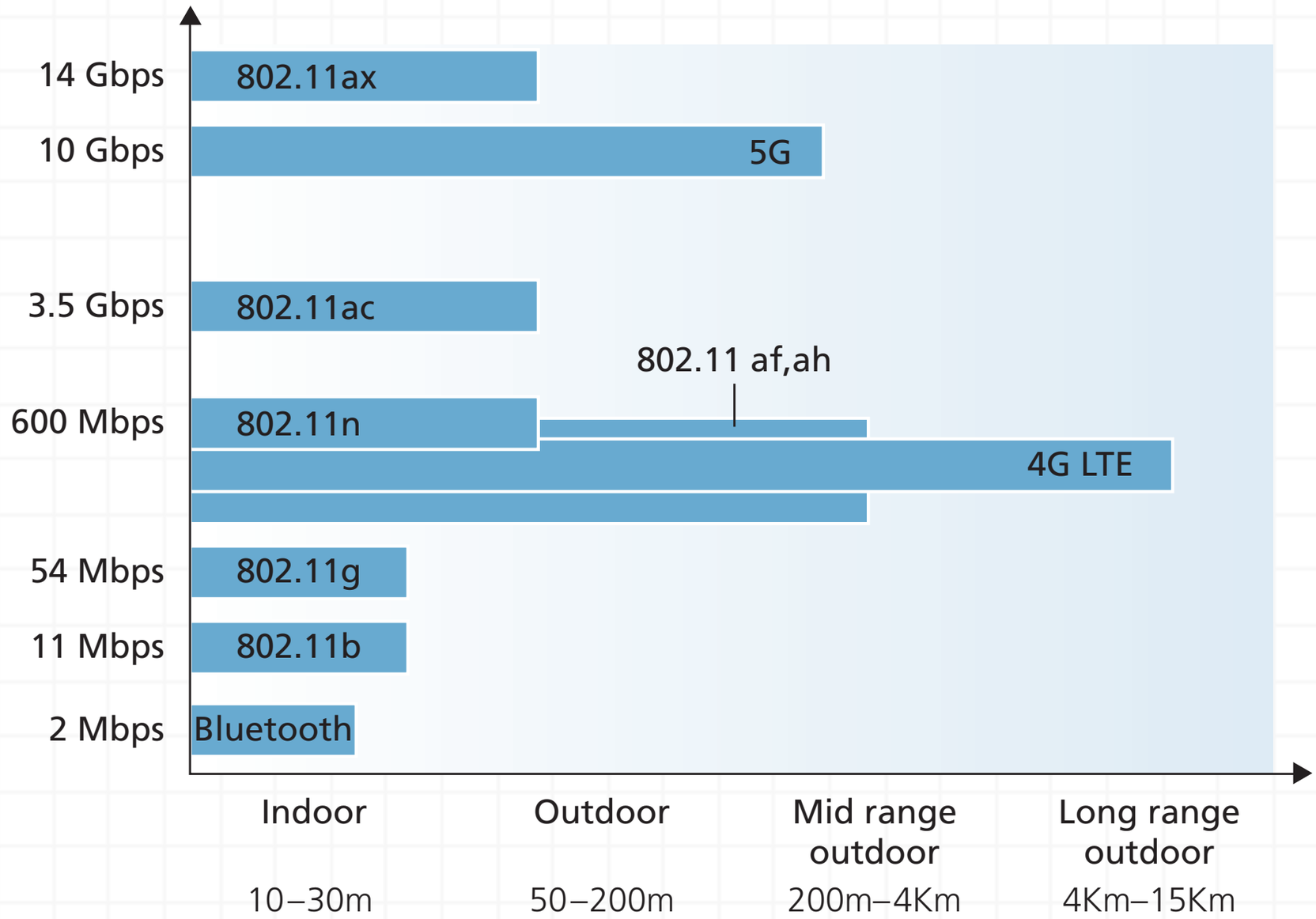
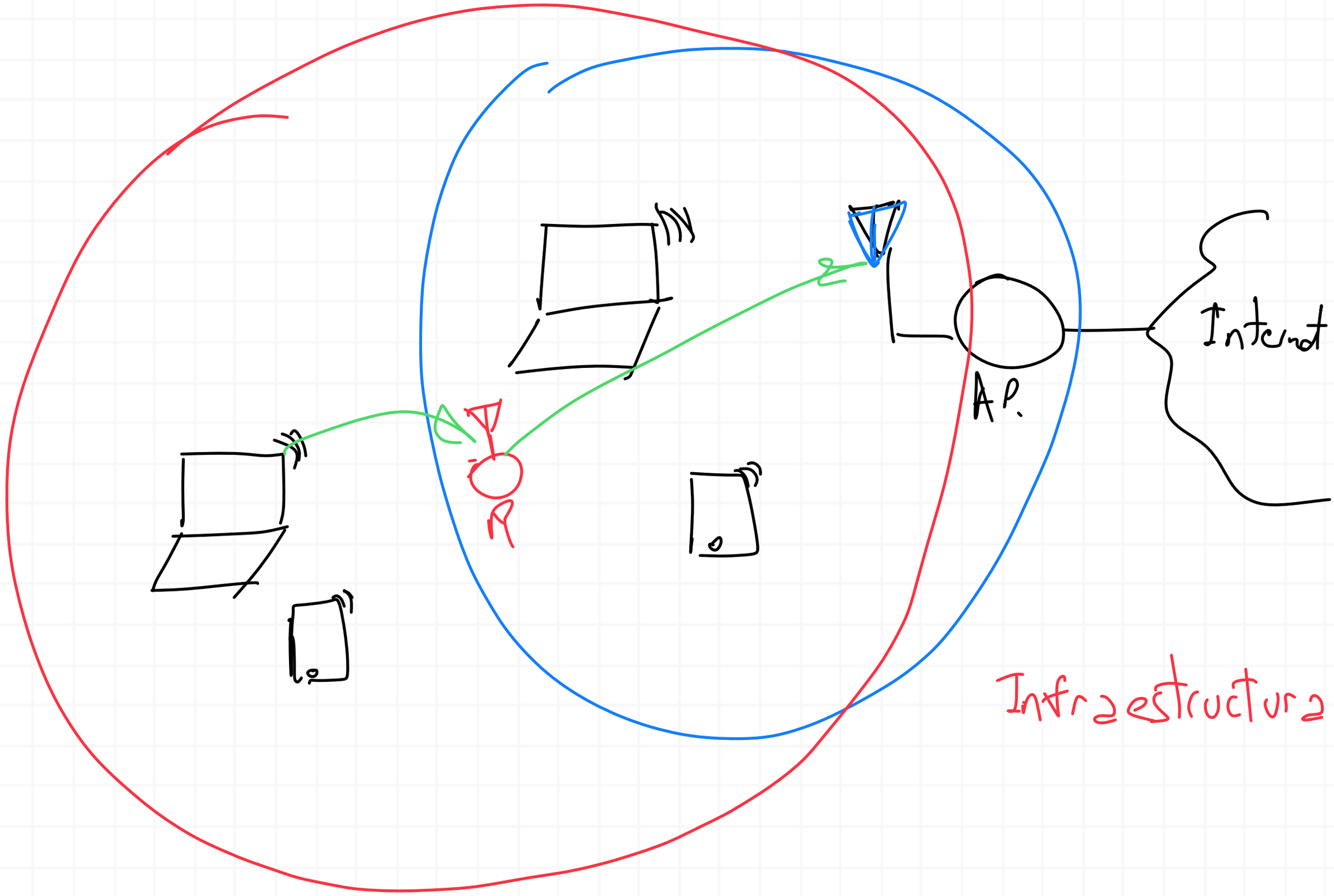
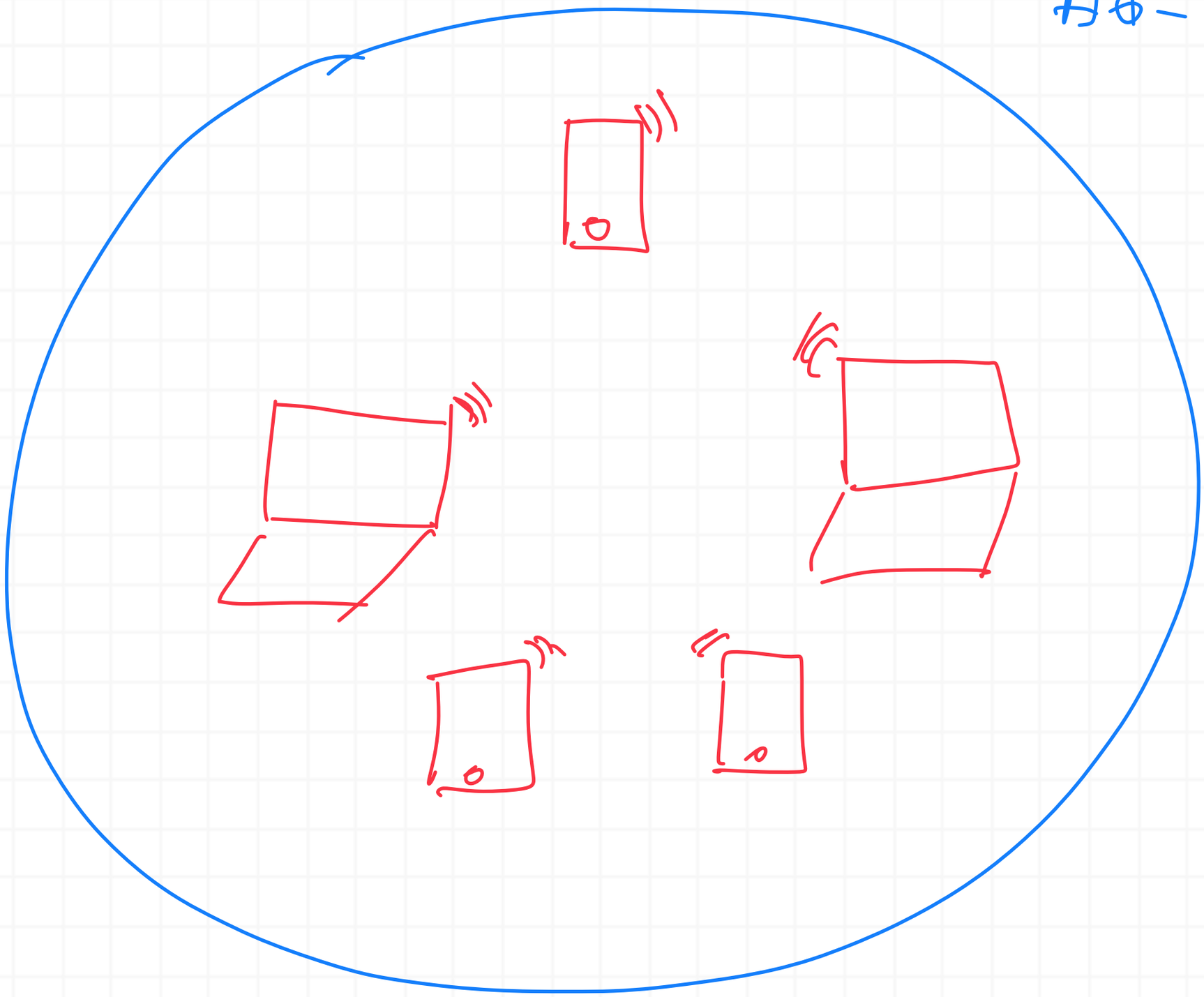


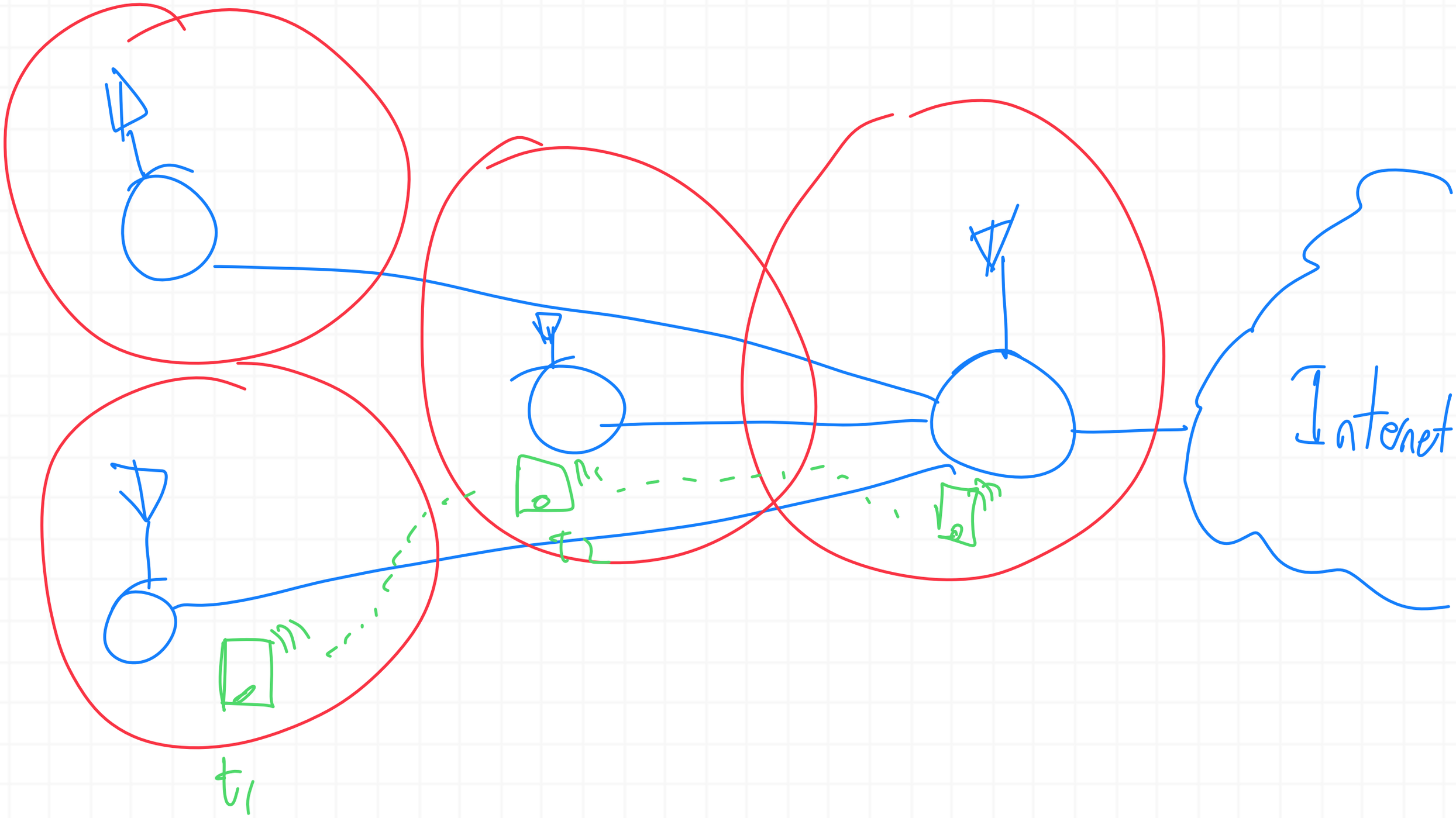
Figure 7.2 ♦ Wireless transmission rates and range for WiFi, cellular 4G/5G and Bluetooth standards (note: axes are not linear)

[James Kurose and Keith Ross. Publisher: Pearson, Edition: 8th, 2021]


A.P.
Internet
Infraestructura

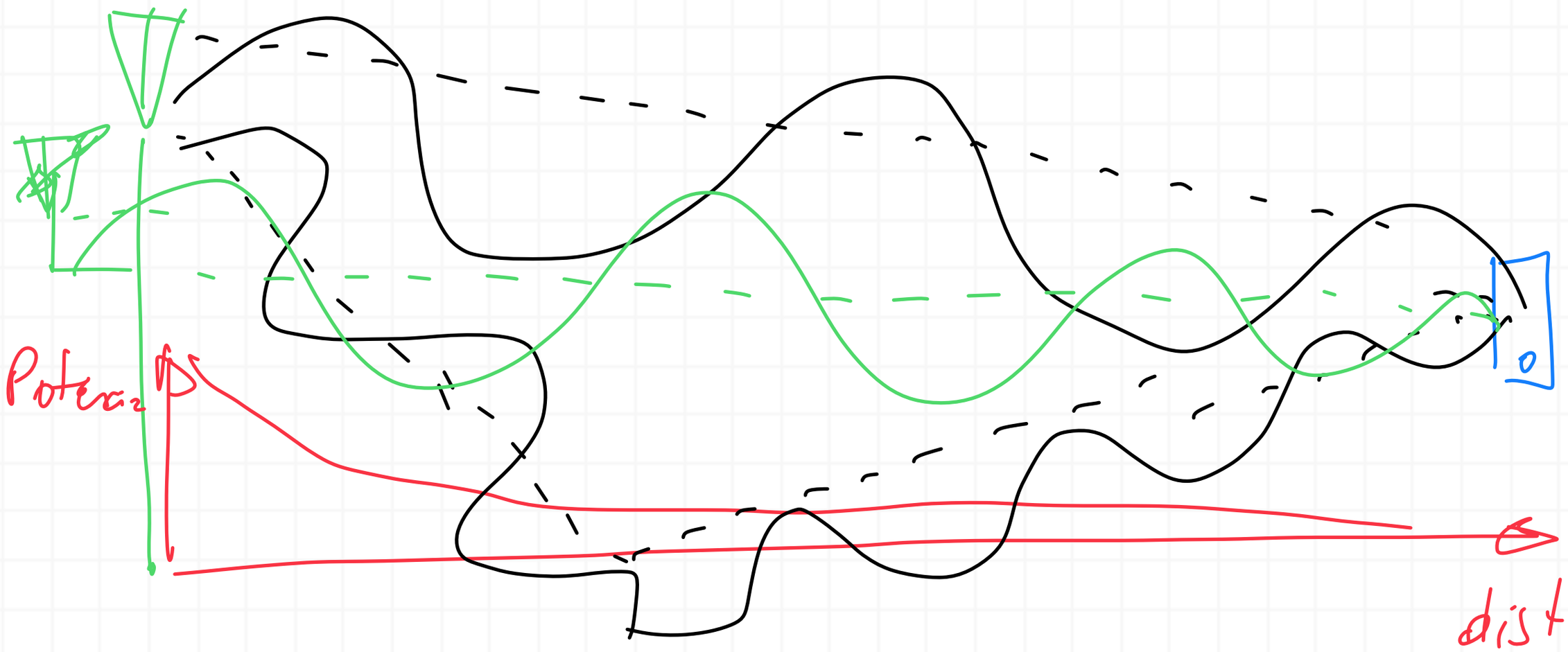
Ad-hoc




Internet
t_1

Características físicas de los medios

- Difusión, alcance (emisores escondidos)
- Efectos adversos (*impariments*): atenuación, interferencia (múltiples caminos, otras fuentes)
- Mediciones: relación señal a ruido (*SNR, signal-to-noise*) y tasa de bits erróneos (*BER, Bit Error Rate*)
- Dependencia del tipo de modulación y el BER.



$$SNR = \frac{S\,[W]}{N\,[W]}$$

$$BER = \frac{\text{bit errados}}{\text{bits transmitidos}}$$

$$F.O. = 10^{-9}$$

Wireless $10^{-3} \approx 10^{-2}$

$\hookrightarrow$ FEC $\approx 10^{-9}$

$$[dB] = 10 \log_{10} \frac{P_1}{P_2}$$

$$SNR_{dB} = 10 \log_{10} \frac{S[W]}{N[W]}$$

$10\,dB \Rightarrow 10^{\frac{10}{10}} = 10$ veces

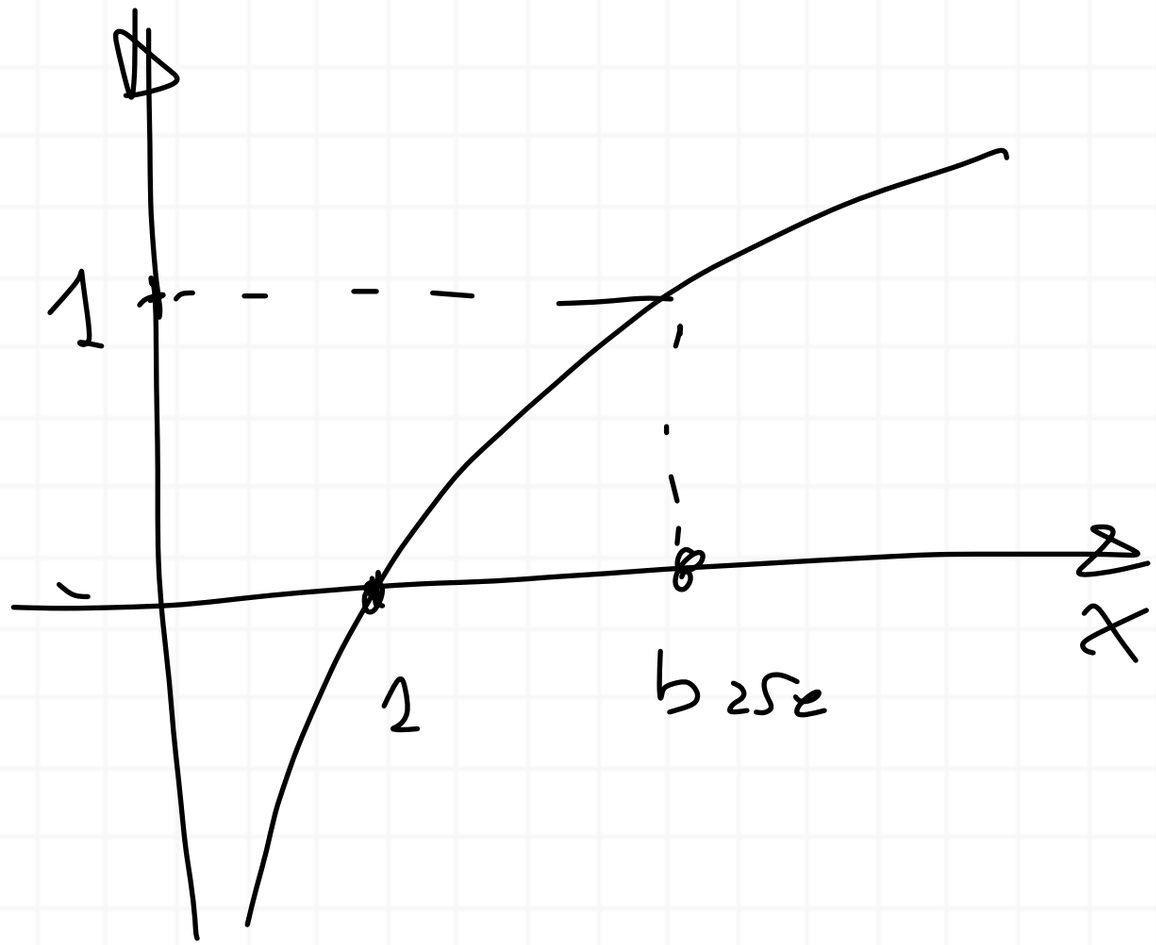
$20\,dB \Rightarrow 10^{\frac{20}{10}} = 100$ veces

$30\,dB \Rightarrow 10^{\frac{30}{10}} = 1000$ veces

$S = -79\,dB_m$ $\quad N = -90\,dB_m$

$-79 - (-90) = +11\,dB$

$\log_{base} x$



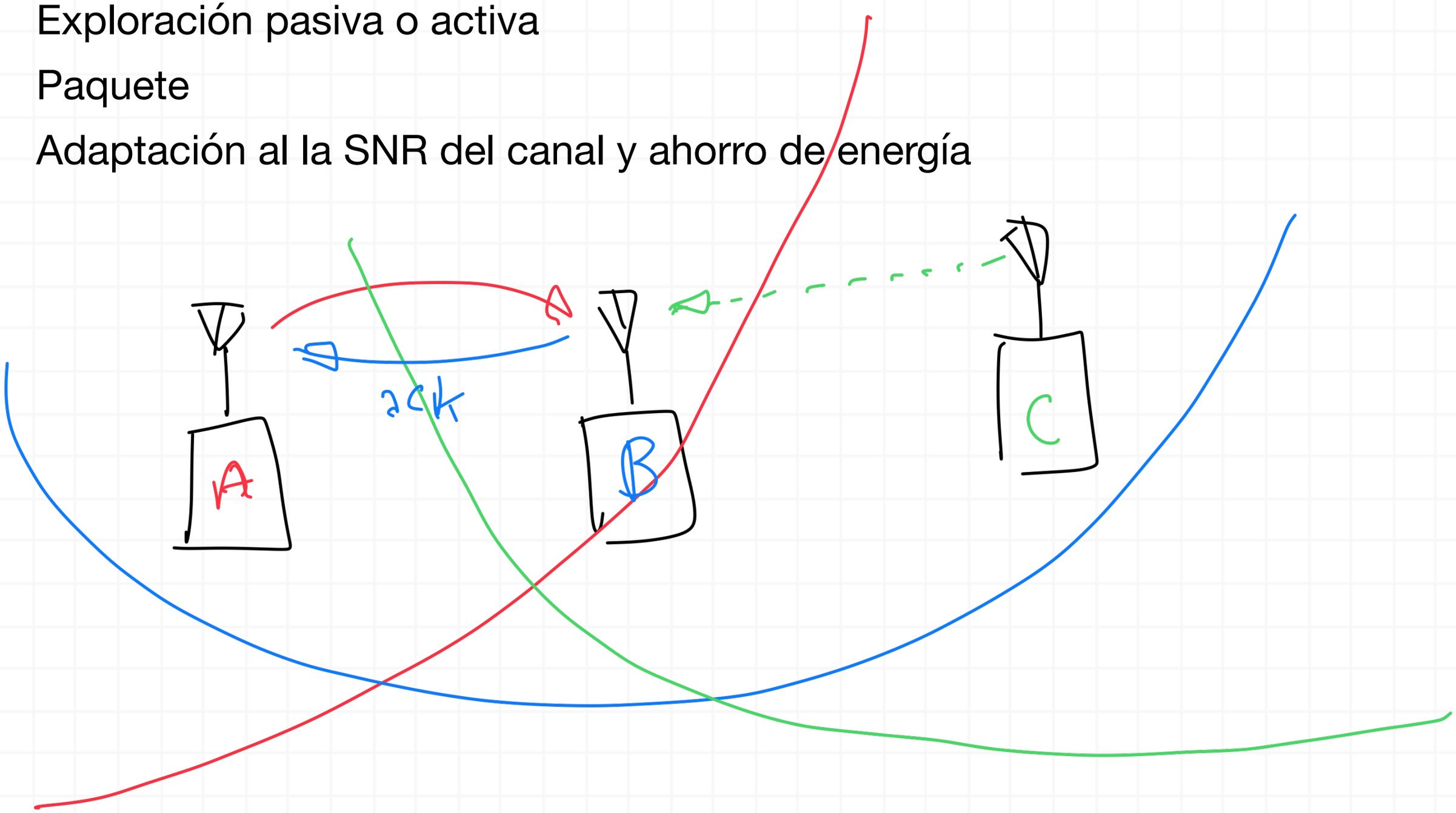
$dBm = 10 \log_{10} \frac{P}{1mW}$

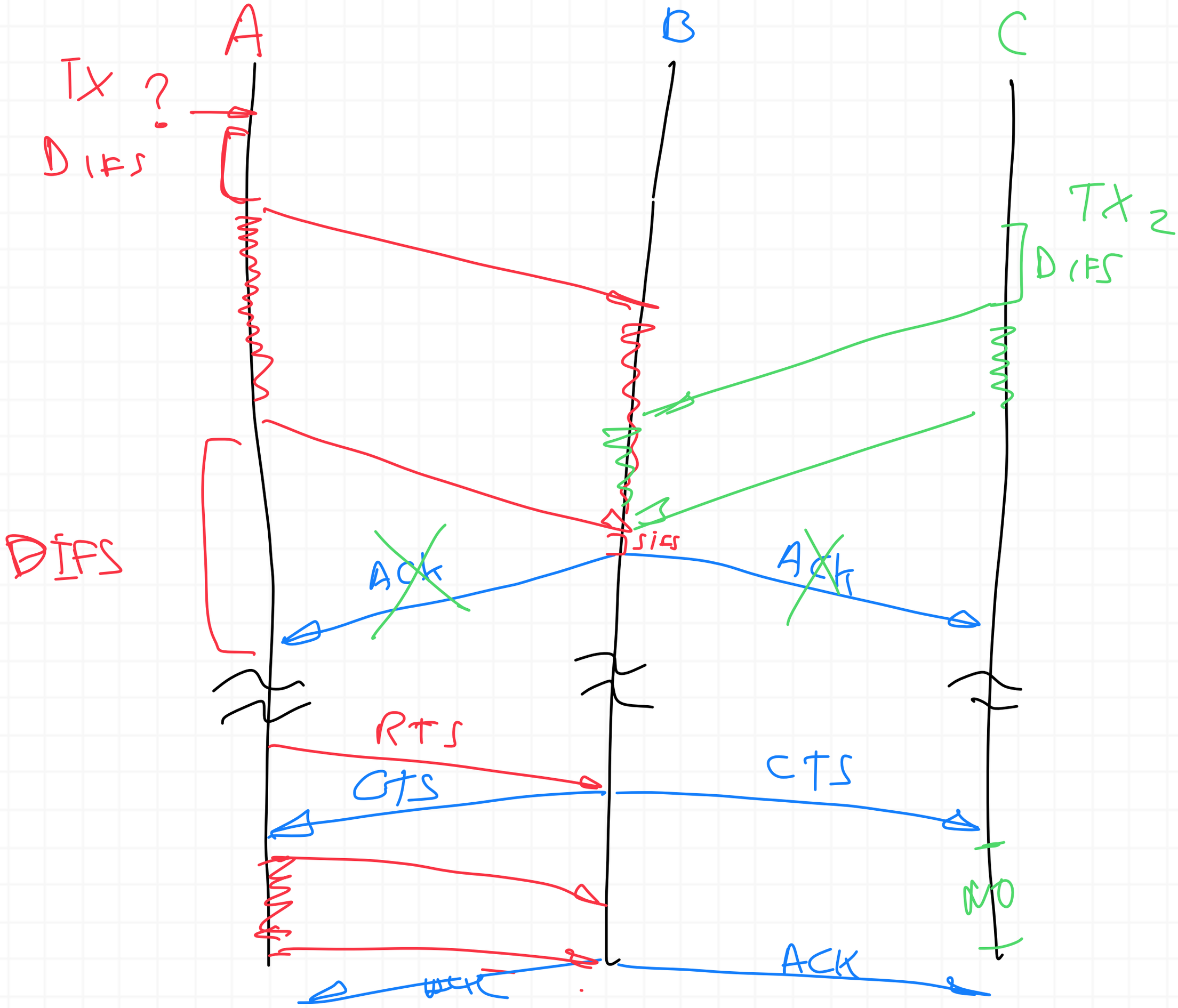
$10\,dBm - 30\,dB = -20\,dBm$

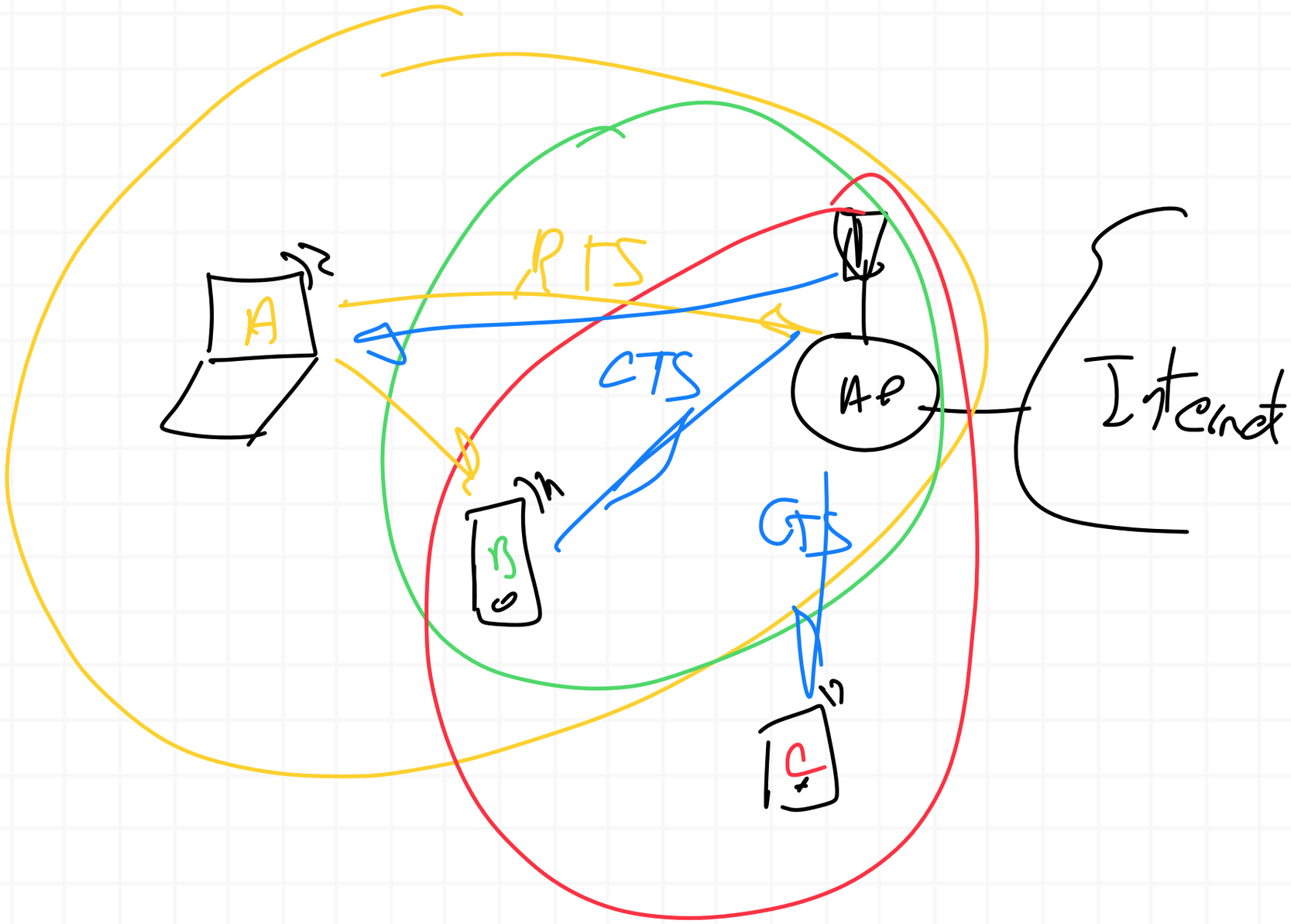
$10\,mW \qquad\qquad 10\,\mu W$

WiFi: IEEE 802.11

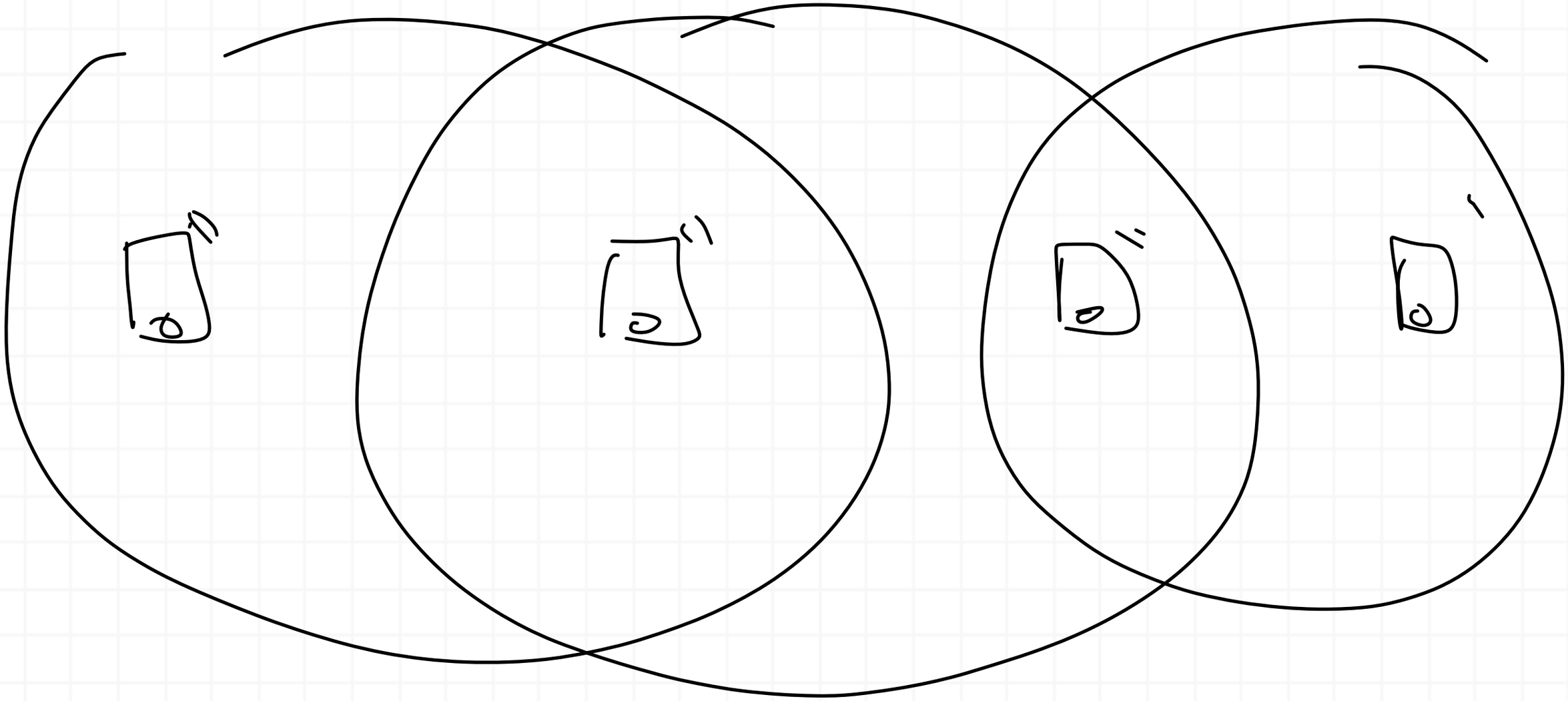
- Versiones a (54 Mbps, 5 GHz), b (11 Mbps, 2.4 GHz), g (54 Mbps, 2.4 GHz) n (450 Mbps, 2.4 GHz y 5 GHz), ac (1300 Mbps, 5 GHz)
- Acceso al medio (CSMA/CA) con y sin RTS/CST
- Arquitectura (AP, repetidor, ad-hoc)
- Exploración pasiva o activa
- Paquete
- Adaptación al la SNR del canal y ahorro de energía



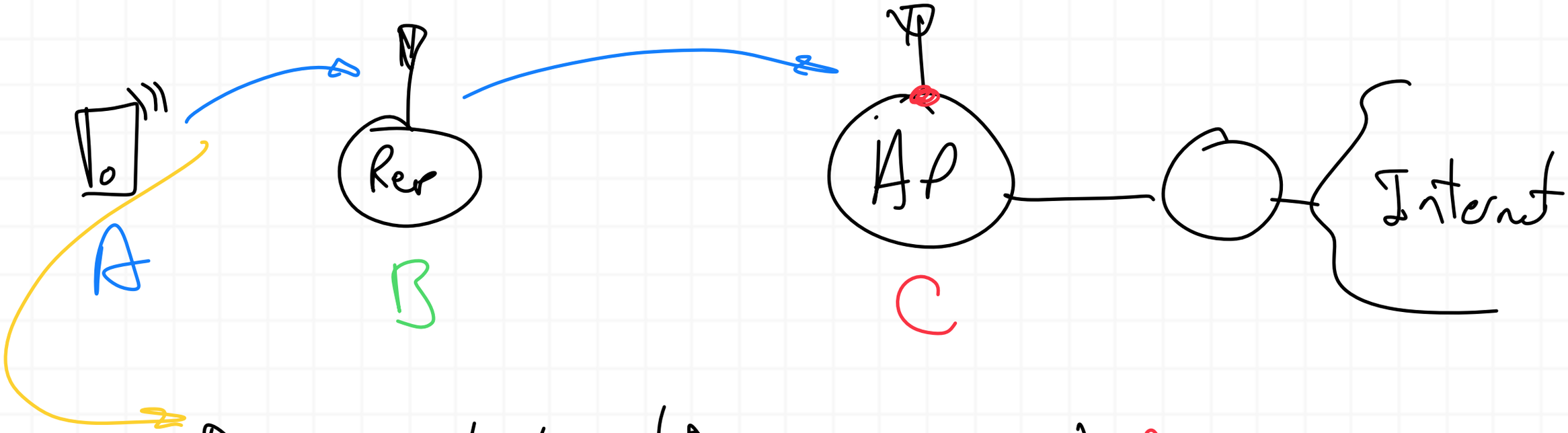

A
B
C
TX ?
DIFS
TX 2
DIFS
DIFS
SIFS
ACK
ACK
RTS
CTS
CTS
NO
ACK


A
RTS
CTS
AP
Internet
B
CTS
C

Ad-hoc



~~RTS/RTS~~



Dirección destino | Dirección Origen | Dirección del Router

B A C

IEEE 802.11 Velocidad = f(SNR)

Lectura para la próxima clase:
leer desde el Capítulo *7.4 Cellular Internet Access, hasta el final.*